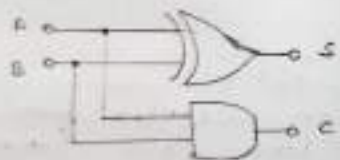


Apparatus Required

S.R. No.	Apparatus	Specification	Quantity
1	Integrated chips (IC)	NAND gate - 7401 AND gate - 7408 OR gate - 7432 X-OR gate - 7406 etc.	1 1 1 1
2	Connecting wires & power supply	-	-
3	Digital trainer kit	-	1
4	Logic probe	-	1



(Half-adder logic diagram)

Object:- To study & realize half & full adder.
 (i) Using Basic gates
 (ii) Using Only NAND gates.

Theory:-

Half-adder:-

The simplest combinational circuit which performs the arithmetic addition of two binary digits is called Half-adder. A half adder adds two one-bit binary numbers A & B. It has two outputs, S & C (the value theoretically carried on to the next addition); the final sum is $2C + S$.

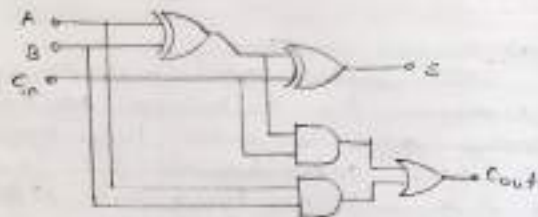
From the truth table, the logic expression for the sum output can be written as a Sum of Product expression by summing up the input combinations for which the sum is equal to 1. In the truth table, the sum output is 1 when $AB = 01$ & $AB = 10$. Therefore, the expression for sum is

$$S = \bar{A}B + A\bar{B}$$

Now, this expression can be simplified as

$$S = A \oplus B$$

Similarly, the logic expression for carry output can be expressed as a sum of Product expression



* Full Adder Logic Diagram *

by summing up the input combination for which the carry is equal to 1 in the truth table, the carry is 1 when, $AB = 11$. Therefore,
 $C = AB$

Full-Adder:- A Full-adder is a combinational circuit that performs the arithmetic sum of three input bits and produces a sum output and a carry.

From the truth table, the logic expression for S can be written by summing up the input combination for which the sum output is 1 as:

$$S = \bar{A}\bar{B}C_{in} + \bar{A}B\bar{C}_{in} + A\bar{B}\bar{C}_{in} + ABC_{in}$$

$$S = \bar{A}(\bar{B}C_{in}) + A(B\bar{C}_{in})$$

$$\text{Let } B\bar{C}_{in} = X$$

$$S = \bar{A}X + AX = A \oplus X$$

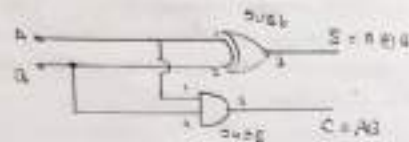
So

$$S = A \oplus B \oplus C_{in}$$

Similarly, the logic expression for C_{out} can be written by summing up the input combinations for which C_{out} is 1, as given below:

$$C_{out} = BC_{in} + A\bar{B}C_{in} + A\bar{B}\bar{C}_{in}$$

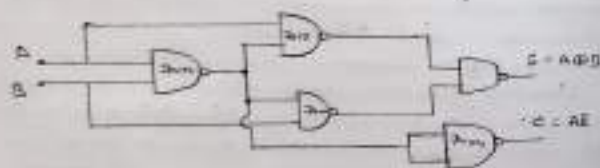
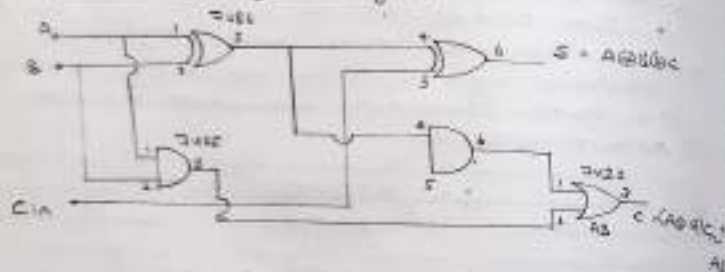
Half adder using Basic gates-



$$S = A \oplus B$$

$$C = AB$$

Full adder using basic gates



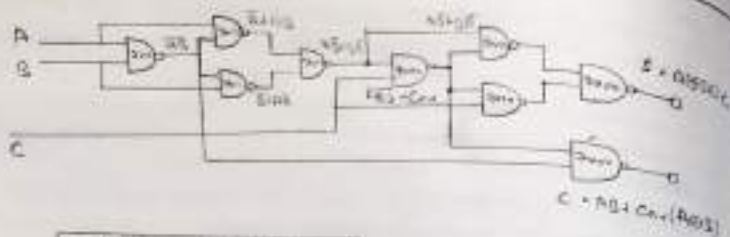
Half Adder

Now, ABC_{in} term is added twice for simplification

$$C_{out} = BC_{in} + A\bar{B}C_{in} + AB\bar{C}_{in} + ABC_{in} + ABC_{in}$$

$$C_{out} = BC_{in} + B\bar{C}_{in} + AB$$

Using NAND gates Half adder Circuit Diagram



Input		Output	
A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Half adder

Input			Output	
A	B	C _{in}	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Full adder

Observation table :-

Procedure :-

- 1) Verify the gates.
- 2) Make the connections as per the circuit diagram.
- 3) Make pin no. 7 grounded to connect pin no. 14 to power supply.
- 4) Switch on Vcc to apply various combinations of input according to the truth table.
- 5) Note down the output reading for half/full adder sum/difference and the carry/borrow bit for different combinations of inputs.

Precautions :-

1. Insert IC on board tightly.
2. Don't forget to ground pin no. 7.
3. Hold the IC properly.

Result :- Study & realization of half adder & full-adder circuits have been done using IC's 7410, 7402, 7432 etc.

Discussion :- Using these IC's we can realize all & half adder & other combinational circuits easily. Since NAND is universal gate so we generally use IC 7400.

Teacher's Signature

Viva Question

Q1 What is an adder?

Ans An adder is a combinational logic circuit used to perform the addition of binary numbers.

Q2 What are the types?

Ans There are two main types of adder -

(1) Half adder (2) Full adder.

Q3 What is half adder?

Ans A half adder is a combinational circuit that adds two single-bit binary numbers to produce two output sum & carry.

Q4 What are the inputs & outputs of a Half adder?

Ans Input A, B

Output :- Sum (S), Carry (C)

Q5 Write the Boolean expressions for the half adder?

Ans Sum = $A \oplus B \Rightarrow$ XOR Operation

Carry = $A \cdot B \Rightarrow$ AND Operation

Q6 Write the Boolean expression for full adder with input & output.

Ans Input $\rightarrow A, B, C_{in}$, Output $\Rightarrow$ Carry (Count), Sum (S)

Sum = $A \oplus B \oplus C$

Carry = $(A + B) \cdot C_{in} + A \cdot B$